

Machine Learning Course Project Report

Text Classification on AG News

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1 Course Context

- Course: Machine Learning (CO3117)
- Department: Computer Science and Engineering, HCMUT – VNU-HCM
- Supervisor: Dr. Le Thanh Sach
- Topic: Text Data Classification (Topic 2)

2 Project Objectives

The project aims to build a complete text classification pipeline from exploratory data analysis to final model comparison, following the required workflow:

1. EDA and data understanding.
2. Configurable preprocessing and TF-IDF feature extraction.
3. Classical ML training and evaluation (Logistic Regression, SVM, Naive Bayes).
4. Modern embedding extension with SBERT.
5. Comparison across feature families with report-ready artifacts.

3 Dataset and Problem Setting

- Dataset: `ag_news` from Hugging Face Datasets.
- Task: 4-class news topic classification.
- Split used in full mode: 120,000 training samples and 7,600 test samples.

4 Source Code and Reproducibility

The complete source code, artifacts, and instructions to run this project are available on our public GitHub repository:

https://github.com/THVKhang/MachineLearning_TextModule

All executions are designed to be reproducible via a single Colab Notebook entry point, without requiring personal cloud storage mounting.

5 Exploratory Data Analysis (EDA)

Extensive Exploratory Data Analysis was performed to understand the characteristics of the AG News dataset. Key findings include:

- **Balanced dataset:** The 120,000 training samples are perfectly balanced across the 4 classes (30,000 each). This eliminates the need for class weighting or resampling techniques.
- **Short texts:** The average text length is approximately 35 words. Most semantic content can be captured effectively using TF-IDF with bigrams.
- **Domain-specific vocabulary:** Each class has distinct top keywords (e.g., “game” for Sports, “stock” for Business).
- **Low noise:** The dataset is very clean, with near-zero empty or duplicate texts.
- **Minimal URLs:** Only about 32 URL-heavy texts were found in the training set, having negligible impact on overall quality.

These findings motivated our use of standard TF-IDF and SBERT models without the need for complex outlier handling.

6 Implemented Methodology

6.1 Traditional Pipeline

- Preprocessing options: lowercasing, stopword removal, punctuation removal, number removal.
- TF-IDF configuration search over three candidates:
 - `tfidf_uni_1k`
 - `tfidf_uni_bi_3k`
 - `tfidf_uni_bi_5k`
- Primary model-selection metric: F1-weighted.
- Trained models: Logistic Regression, SVM, Naive Bayes.

6.2 Modern Embedding Extension

- Embedding model: `sentence-transformers/all-MiniLM-L6-v2`.
- Benchmark scales: `5k_2k` and `20k_2k`.
- Embeddings and labels are exported to `.numpy` artifacts.

7 Engineering Workflow Integration

A one-entry workflow was implemented to improve reproducibility:

- Planner chooses objective-specific strategy (fast, balanced, best).
- Runners execute TF-IDF and embedding branches.
- Critic validates quality thresholds, required scales, and artifacts.
- Reporter generates concise recommendation outputs.

8 Results

8.1 Best TF-IDF Results (Full Data)

Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
SVM	0.9046	0.9044	0.9046	0.9044
Logistic Regression	0.9038	0.9036	0.9038	0.9036
Naive Bayes	0.8871	0.8866	0.8871	0.8867

Table 1: TF-IDF model comparison from `reports/tables/tfidf_model_comparison.csv`.

8.2 SBERT Benchmark Results

Scale	Best Model	Accuracy	Precision (w)	Recall (w)	F1 (w)
5k_2k	Logistic Regression	0.8740	0.8734	0.8740	0.8736
20k_2k	SVM	0.8955	0.8958	0.8955	0.8954

Table 2: Best embedding model per scale from `reports/tables/bert_benchmark_summary.csv`.

8.3 Feature-Family Comparison

The final comparison between best TF-IDF and best embedding model shows:

- Best TF-IDF: SVM with $F1\text{-weighted} = 0.9044$.
- Best embedding: SVM at `20k_2k` with $F1 = 0.8954$.

Therefore, TF-IDF + SVM is selected as the final recommendation for this dataset and setup.

9 Artifact Outputs

The project exports reproducible artifacts under **features/** and **reports/**, including:

- Feature arrays (TF-IDF and SBERT) in **.npy** format.
- Confusion matrices for all TF-IDF models.
- Benchmark and comparison CSV tables.
- Workflow logs, critic report, and reporter summary.

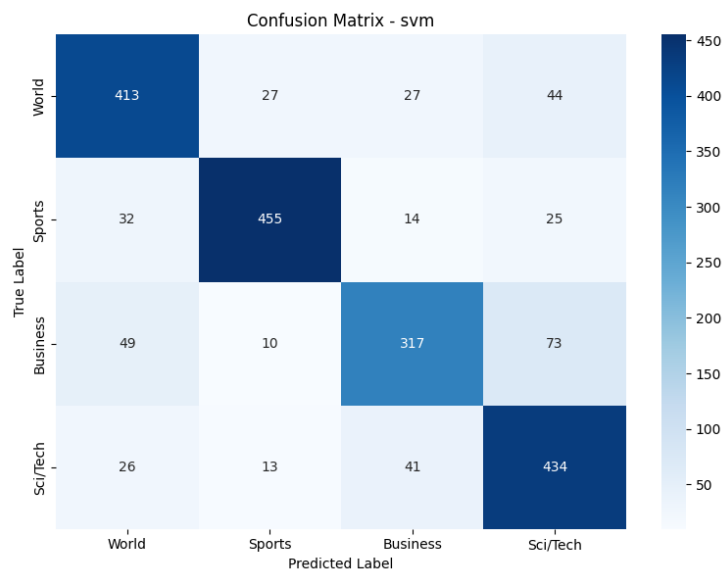


Figure 1: Confusion matrix of best TF-IDF model (SVM).

10 Team Contributions

Member	ID	Main Contribution	%
Tran Hoang Vy Khang	2352502	Embedding benchmark, integrated workflow, final comparison and outputs	30
Le Tan Minh Khoa	2352563	Text cleaning and TF-IDF feature engineering	20
Tao Nguyen Quang Khang	2352499	Classical model training and evaluation on TF-IDF	20
Tran Gia Huy	2252264	Pipeline/configuration support and reproducibility structure	15
Vo Le Hai Dang	2352257	Data loading and EDA notebook analysis	15

Table 3: Detailed contribution table (sum = 100%).

11 Conclusion

The required course targets were completed:

- Complete traditional ML pipeline implemented and validated.
- Modern embedding extension benchmarked at multiple scales.
- Direct TF-IDF vs embedding comparison produced.
- Automated planner-runner-critic-reporter workflow built for reproducibility.

Final recommendation: use TF-IDF with SVM as the primary baseline for AG News in this project.